Proposed Solution Template

Date	12-06-2026
Team ID	
Project Name	**Voice Based Concept Understanding Analyser**
Maximum Marks	2 Marks

Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Traditional oral assessments are subjective, time-consuming, and difficult to evaluate consistently. Teachers face challenges in assessing students' conceptual understanding, speech clarity, and communication skills while providing personalized feedback.
2.	Idea / Solution description	Develop an AI-powered Voice-Based Concept Understanding Analyser (VBCUA) that evaluates students' spoken explanations by converting speech to text, analyzing semantic similarity with reference answers, assessing speech fluency, and generating personalized feedback with downloadable PDF reports.
3.	Novelty / Uniqueness	The system combines speech recognition, semantic analysis, audio feature extraction, and generative AI in a single platform to evaluate both conceptual understanding and communication skills, offering instant, objective, and personalized assessment.
4.	Social Impact / Customer Satisfaction	The solution promotes fair and unbiased evaluations, enhances students' communication and conceptual learning, reduces teachers' workload, and improves the overall learning experience through immediate AI-driven feedback.
5.	Business Model (Revenue Model)	The platform can be offered as a Software-as-a-Service (Sa institutions, with premium plans for advanced analytics, cu integration, and enterprise deployment.
6.	Scalability of the Solution	The system is cloud-ready and can be scaled to support multiple institutions, thousands of concurrent users, multilingual assessments, integration with Learning Management Systems (LMS), and additional AI-based evaluation modules in the future.